



POLITECNICO
MILANO 1863

Visual Reconstruction of Played Music

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I. PROBLEM STATEMENT

The objective of this project is to analyze a video of a played piano keyboard in order to identify the pressed keys at each instant. Given a monocular RGB video, the task is decomposed into multiple stages, including keyboard localization, identifying individual keys, tracking fingertip positions, and determining which keys are pressed over time.

The system is required to operate on standard video without any specialized sensors. All detections must be expressed in the full image coordinate system to allow interaction between different processing stages.

The main challenge arises from the presence of hands, reflections, and background elements, which make direct analysis on the full image difficult. For this reason, the processing pipeline is structured into multiple stages, where intermediate representations are used to simplify subsequent steps.

II. METHOD

A. Overall Pipeline

The proposed pipeline processes the input video frame by frame. First, representative frames are selected from the video. Then, the piano keyboard is localized and isolated. White and black keys are detected on the keyboard region. Fingertips are detected on the full image. Finally, fingertip positions are associated with keys to identify pressed keys at each time instant.

B. Frame Selection

From each input video, a subset of frames is extracted. Frames with minimal hand occlusion are selected. These frames are used for keyboard and key detection.

C. Keyboard Localization

The piano keyboard is localized in the selected frames. The keyboard region is isolated from the background to simplify further processing.

D. Piano Key Detection

White and black piano keys are detected inside the keyboard region. White keys are detected first to define the keyboard structure. Black keys are detected relative to the white key layout.

E. Fingertip Detection

Fingertips are detected on the full image. Detection is performed independently from key detection. This ensures that fingertip positions remain consistent with the original image coordinates.

F. Key-Finger Association

Detected fingertip positions are compared with key locations. A fingertip is assigned to a key if it lies inside the key region. Temporal consistency is used to reduce false detections.

G. Pressed-Key Detection

The pressed keys are determined through the combination of spatial and motion-based information. First, the spatial information of the position of the fingertips is mapped to the identified key regions. A key is said to be pressed if the fingertip is spatially associated with the key and the finger corresponding to the key is identified as pressed through articulation analysis.

III. IMPLEMENTATION

A. Pipeline Structure and Intermediate Outputs

The implementation follows a three-stage pipeline. Each stage produces intermediate results that are reused by later stages. All intermediate outputs are saved to disk to allow modular processing and easier debugging. Table I summarizes the main intermediate outputs produced by each stage of the pipeline.

B. Frame Extraction and Hand-Free Frame Selection

Input videos are processed frame by frame. Frames are sampled at a fixed rate. Among the sampled frames, a frame with minimal hand occlusion is selected manually. This frame is used for keyboard and key detection.

TABLE I
PIPELINE STAGES AND SAVED INTERMEDIATE OUTPUTS

Stage	Saved Outputs
Preparation	Hand-free frame, keyboard ROI, masked keyboard image, white and black key coordinates (JSON), key overlay image
Hand analysis	Tracked fingertip positions, finger landmarks, per-finger press information (CSV)
Integration	Fingertip-to-key mapping, pressed keys per frame, final visualization video

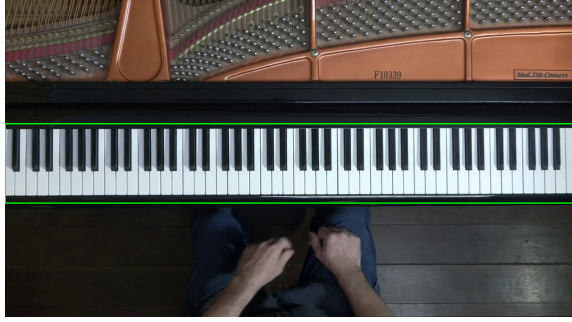


Fig. 1. Estimated keyboard region of interest (ROI) obtained from the hand-free frame.

C. Keyboard Region Localization and Masking

The piano keyboard region is localized using contour-based analysis. A bounding box is estimated to isolate the keyboard from the background. This bounding box defines the region of interest (ROI).

A binary mask is applied to the ROI. Only pixels belonging to the keyboard are preserved. Background regions are removed. This simplifies the subsequent key detection process.

D. White and Black Key Detection

Key detection is performed on the masked and cropped keyboard image. White keys are detected first. Vertical intensity transitions are analyzed to locate borders between adjacent white keys. The median distance between detected borders is used to estimate the average white key width.

Using this width, a regular grid of white key bounding boxes is generated. The height of white keys is estimated from the keyboard region dimensions and kept constant across the keyboard.

Black keys are detected relative to the white key layout. Only valid musical positions are considered. Detection is restricted to the upper portion of the white keys, where black keys are expected to appear.

All detected key coordinates are initially expressed in the cropped image reference frame. They are then mapped back to the full image reference frame using the known ROI position and scale. The final key locations are stored as JSON files.

E. Hand Motion Analysis and Press Action Detection

The goal of this section is to analyse hand motion during piano performance and extract the information required to

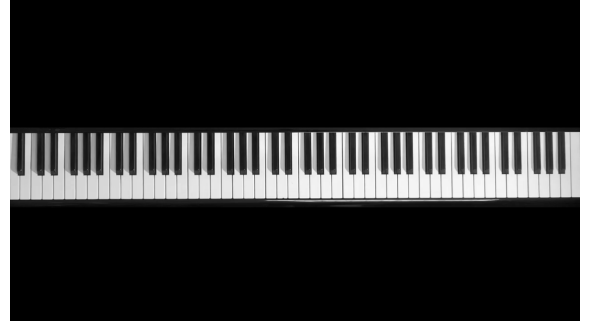


Fig. 2. Masked keyboard image after background removal.

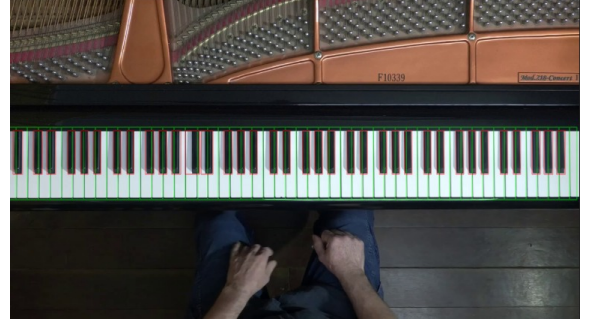


Fig. 3. Detected white and black piano keys overlaid on the full image.

infer key pressing events. Unlike keyboard geometry estimation, which relies mainly on static image features, hand analysis must operate reliably under dynamic conditions such as rapid motion, partial occlusions, and varying finger poses. The proposed approach therefore combines deep-learning-based hand landmark detection with geometric analysis of finger articulation and temporal filtering.

The module produces two main outputs:

- Fingertip coordinates for each frame
- A binary press state for each finger over time.

1) *Fingertip Detection*: Reliable fingertip localization is a fundamental prerequisite for determining the interaction between fingers and keys. Classical contour-based approaches based on convex hulls or convexity defects were considered but found to be sensitive to segmentation noise and finger overlap. For this reason, a learning-based hand landmark detection approach was adopted.

We employ MediaPipe Hands, a pretrained hand pose estimation framework capable of predicting 21 landmarks for each detected hand in image coordinates. The fingertips are identified by selecting the landmark indices corresponding to the distal points of the thumb, index, middle, ring and little fingers.

The processing pipeline for each frame consists of:

- 1) detecting the hand region
- 2) estimating the full set of hand landmarks
- 3) extracting fingertip coordinates
- 4) storing the results in a structured CSV representation.

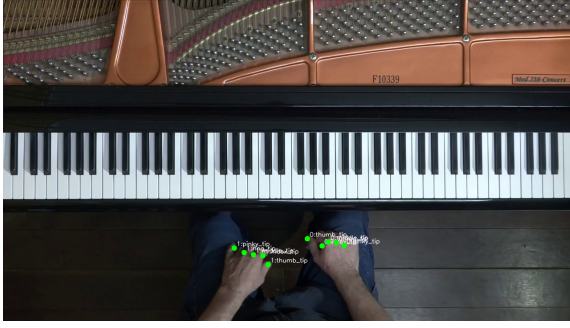


Fig. 4. Example frame showing detected hand landmarks and fingertip positions obtained using MediaPipe Hands.

Figure 4 illustrates an example of the detected hand landmarks and fingertip positions.

2) *Temporal Tracking and Refinement*: Since landmark detection is performed individually per frame, the output of raw detection may vary or skip due to motion blur or self-occlusion. In order to keep things stable, the fingertip paths are further refined with temporal consistency constraints. In detail:

- detection is connected between consecutive frames
- outliers are removed
- small detection gaps are filled with temporal filtering

This provides smooth fingertip paths, which are important for stable motion-based analysis in the following steps.

3) *Press Action Detection*: The position of the fingertips is not enough to identify the press of a key as the fingers can be moved without pressing keys laterally or hovers above without applying pressure. Thus, press detection is posed as a study of finger articulation. The point to note here is that striking a piano key usually causes a typical bending of the finger. The geometric features are calculated using the complete set of hand landmarks and are measured. Inter-finger distances between pairs of joints picked. A decrease of these distances is a sign of increased finger flexion.

For each frame and for each finger:

- 1) joint-distance features are calculated
- 2) based on these distances, there are bending metrics
- 3) The state of the press is determined by the threshold-based classification

The thresholds were chosen empirically to compromise between sensitivity to real presses and robustness against normal hand motion.

4) *Temporal Stabilization of Press Decisions*: Instantaneous press decisions may vary due to noise in landmark detection and/or the presence of motion artefacts. Temporal smoothing is carried out on the binary press decisions. In this step, short-term temporal consistency is ensured by:

- Removing single frame decisions
- stabilizing short press intervals
- reducing flickering between pressed and non-pressed states.

The press-state sequence now provides a stable sequence of finger activities, which will eventually be used with keyboard geometry and finger-to-key associations to determine the pressed keys.

F. Key-Finger Mapping

For each frame, fingertip positions were compared with detected key bounding boxes. A fingertip was assigned to a key if it was located inside the corresponding key region. Multiple fingertip detections were handled independently.

G. Pressed-Key Detection

Key press events were detected over time. A key was considered pressed when a fingertip remained inside its region across consecutive frames. Short temporal smoothing was applied to reduce spurious detections.

IV. EXPERIMENTS AND RESULTS

A. Dataset

The system was evaluated on multiple piano-playing videos. All videos were recorded using a static camera. Different video resolutions and quality levels were used. Both high-resolution and low-resolution videos were included. Some videos contained blur and compression artifacts. This was done to observe how the pipeline behaves under different visual conditions.

B. Qualitative Evaluation

Intermediate results were evaluated visually. The keyboard ROI, masked keyboard image, and key detection overlays were inspected. Detected white and black keys showed good alignment with the actual keyboard layout.

C. Integration Results

Detected key locations were combined with fingertip detection results. Since all outputs were expressed in full image coordinates, key-finger mapping was performed directly. Pressed keys were correctly identified in representative piano sequences.

D. Limitations

The method assumes a static camera and a fixed keyboard position. Strong reflections or severe occlusions may affect detection results. These cases were not addressed in this project.

V. CONCLUSION

A vision-based system for analyzing piano playing from video was implemented. The keyboard region was localized, piano keys were detected, and fingertip positions were associated with keys. Pressed keys were identified over time.

The modular structure of the pipeline allows each processing stage to operate in a consistent coordinate system. Future work may include handling camera motion and improving performance under challenging lighting.